

Reviewed Preprint

v2 • November 11, 2024

Revised by authors

Reviewed Preprint

v1 • August 2, 2024

Kynurenine monooxygenase blockade reduces endometriosis-like lesions, improves visceral hyperalgesia, and rescues mice from a negative behavioural phenotype in experimental endometriosis

Ben Higgins, Ioannis Simitsidellis, Xiaozhong Zheng, Frances Collins, Natalie ZM Homer, Scott G Denham, Joanna P Simpson, Mike Millar, Lyndsey Boswell, Hee Y Lee, Yeon G Kim, Kyung H Park, Larry C Park, Patrick J Sweeney, Gerard Feraille, Alessandro Taddei, David Chagras, Thierry Alvarez, Scott P Webster, Andrew Horne, Philippa TK Saunders, Damian J Mole 

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK • EXPPECT Edinburgh, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK • MRC Centre for Reproductive Health, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK • Mass Spectrometry Core, Edinburgh Clinical Research Facility, The University of Edinburgh, Edinburgh, UK • Centre for Cardiovascular Science, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK • SuRF Molecular Histology Facility, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK • Naason Science, Inc., Cheongju, Republic of Korea • Syneos Health France, Biot, France • Clinical Surgery, The University of Edinburgh, Edinburgh, UK

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **useful** study holds importance within the focused scope of endometriosis treatment, providing initial evidence of a potential new therapeutic target. The strength of the evidence is **solid**, as the methods, data, and analyses support the authors' conclusions regarding the specific aims. The study provides promising preliminary evidence of KMO implication in endometriosis, but it falls short of establishing a strong rationale for proposing KNS898 as a treatment for endometriosis given the limitations in evidence and mechanistic insights.

<https://doi.org/10.7554/eLife.99226.2.sa3>

Abstract

Summary

Endometriosis is a common and debilitating neuro-inflammatory disorder that is associated with chronic pain. Definitive diagnosis is based on the presence of endometrial-like tissue (lesions) in sites outside the uterus. Kynurenine monooxygenase (KMO) is a mitochondrial enzyme of tryptophan metabolism that regulates inflammation and immunity. Here, we show that KMO is expressed in epithelial cells in human endometriosis tissue lesions and in corresponding lesions in a mouse model of endometriosis. In mice, oral treatment with the potent KMO inhibitor KNS898 induced a biochemical state of KMO blockade with accumulation of kynurenine, diversion to kynurenic acid and ablation of 3-hydroxykynurenine production. In the mouse model of endometriosis, KMO inhibition improved histological outcomes and endometriosis pain-like behaviours, even when KNS898 treatment commenced one week after initiation of lesions. Taken together, these results suggest that KMO blockade is a promising new non-hormonal therapeutic modality for endometriosis.

Introduction

Endometriosis is a life-altering condition that affects approximately 10% of females. It is an oestrogen-dependent neuroinflammatory disorder associated with debilitating pelvic pain, excessive fatigue, gastrointestinal and urinary symptoms, and infertility¹. Worldwide, 200 million prevalent cases are forecast by 2026. Endometriosis is defined by the presence of endometrial-like tissue ('lesions') outside the uterus. Physiological hormonal fluctuations in women induce cyclical episodes of cell proliferation, inflammation, injury, and repair within lesions that favour fibroblast to myofibroblast differentiation and fibrosis¹. We and others have identified metabolic dysfunction in cells associated with development of endometriosis lesions². At present, therapeutic options are largely limited to surgery (that often needs to be repeated) or medical therapies that target hormonal activity with resultant side effects (block conception, menopausal symptoms)¹. Patient surveys consistently show frustration with the lack of available treatments that can give long term relief from symptoms including pain, low mood and bloating³. Analysis of recent clinical trials directed at endometriosis¹ has highlighted an unmet need for new, non-hormonal approaches to symptom relief. The studies in the current paper have addressed this need by focusing on an enzyme that is known to play a key role in inflammatory processes that are implicated in the aetiology of endometriosis, but which has not previously been investigated as a target.

Our proposed solution to this unmet medical need is by targeting the enzyme kynurenine 3-monooxygenase (KMO). KMO is a critical regulator of inflammation at multiple organ sites that acts by altering metabolic flux through the kynurenine pathway of tryptophan metabolism⁴. KMO is known to be expressed in non-pathological endometrium⁵, but whether KMO is over-expressed in endometriosis lesions and linked to the severity of inflammation remains to be determined. KMO has been identified as a critical step in converting kynurenine to the cytotoxic metabolite, 3-hydroxykynurenine (3HK), that is an oxidative stressor, causes protein cross-linking, and regulates the immune-metabolic interface⁴. Although there is no specific information about a direct role of KMO in endometriosis, there is evidence of dysregulated tryptophan metabolism in a recent study using a preclinical non-human primate model of endometriosis⁶, and increased kynurenine pathway flux at the immune-metabolic interface between stromal cells and NK immune cells in endometriosis lesions⁷.

At present, there is a scientific rationale for KMO inhibition in endometriosis, but it remains to be shown in preclinical experiments whether KMO inhibition is efficacious in decreasing lesion volume or behavioural symptoms which are used as a surrogate for pain responses in model systems. KNS898 is a highly specific small molecule KMO inhibitor with potential for use by women with endometriosis, based on favourable characteristics for oral development in terms of bioavailability and predicted half-life^{8,9}. KNS898 is a competitive inhibitor of kynurenine substrate at the active site of KMO with a pIC₅₀ of 8.8⁸⁻¹⁰. We propose that KMO inhibition is a novel therapeutic strategy for endometriosis and, if successful, we will make a significant positive impact for women with this painful, disabling condition. The aim of this project was to obtain proof-of-concept for KMO inhibition as a novel therapy for endometriosis. Specifically, we sought to explore the expression of KMO in biobanked human endometrial and endometriosis lesion tissues, confirm target inhibition of KMO by KNS898 in mice, and define the preclinical efficacy of KNS898 in improving clinical features of disease (specifically hyperalgesia and altered cage behaviour) and reducing endometriosis lesion volume in an experimental mouse model of endometriosis.

Results

KMO is expressed in human eutopic endometrium and human endometriosis tissue lesions

To explore whether we could detect variations between expression of KMO in endometrium (eutopic) within the uterus and a variety of lesions obtained from patients, we conducted detailed immunohistochemistry with a primary antibody specific for KMO. On fixed tissue sections of normal human endometrium KMO expression was most striking in epithelial cells lining the glands (**Figure 1a**, insert B) with lower levels in the luminal layer (insert C). Notably expression in the glands was not uniform (**Figure 1a**). KMO was also strongly immunopositive in human peritoneal endometriosis lesions (**Fig. 1d** to **Fig. 1g**), and evidently mostly localised to the epithelial tissues surrounding the distended endometrial gland-like structures (DEGLS) (**Fig. 1e** and **Fig. 1g**). Expression in the stromal compartment appeared variable. In human ovarian endometriosis lesions, KMO was present at low expression levels in the mesothelial layers (**Fig. 1h** and **Fig. 1i**). Duplex immunohistochemistry with cell phenotype markers CD68 (macrophages) did not show KMO colocalising with these immune cells (data not shown).

Oral KNS898 inhibits KMO in mice

Next, we established that oral dosing of KNS898 by gavage in mice resulted in inhibition of KMO. Using n=3 mice per group, we administered KNS898 at 0.01 mg/kg, 5 mg/kg, and 25 mg/kg twice daily (b.d.) in vehicle for seven days, as described. Plasma drug levels and metabolite concentrations are shown in **Figure 2**. KNS898 dosed at 0.01 mg/kg b.d. resulted in a mean ($\pm$ S.E.M.) plasma drug level of 0.18 ± 0.01 ng/mL, 5 mg/kg resulted in 88.8 ± 22.6 μ g/mL, and 25 mg/kg gave 483.9 ± 84.0 μ g/mL. The difference between groups was statistically significant by one-way ANOVA with post hoc Tukey's test ($P = 0.001$) (**Fig. 2a**). KMO blockade with KNS898 was clearly measurable. A backlog in the KMO substrate KYN was evident: KNS898 dosed at 0.01 mg/kg b.d. resulted in a mean ($\pm$ S.E.M.) plasma level of KYN of 339 ± 39 ng/mL, 5 mg/kg resulted in 4940 ± 483 ng/mL, and 25 mg/kg gave 3682 ± 634 ng/mL. The difference between groups was statistically significant by one-way ANOVA with post hoc Tukey's test ($P = 0.001$). The increase in KYN at maximal inhibition was approximately 14-fold compared to the level seen after KNS898 0.01 mg/kg (**Fig. 2b**). Excess KYN was metabolised to KA by kynurenine aminotransferase: KNS898 dosed at 0.01 mg/kg b.d. resulted in a mean ($\pm$ S.E.M.) plasma level of KA of 629 ± 103 ng/mL, 5 mg/kg resulted in 14399 ± 3394 ng/mL, and 25 mg/kg gave 15965 ± 789 ng/mL. The difference between groups was statistically significant by one-way ANOVA with post hoc Tukey's test ($P = 0.001$). The

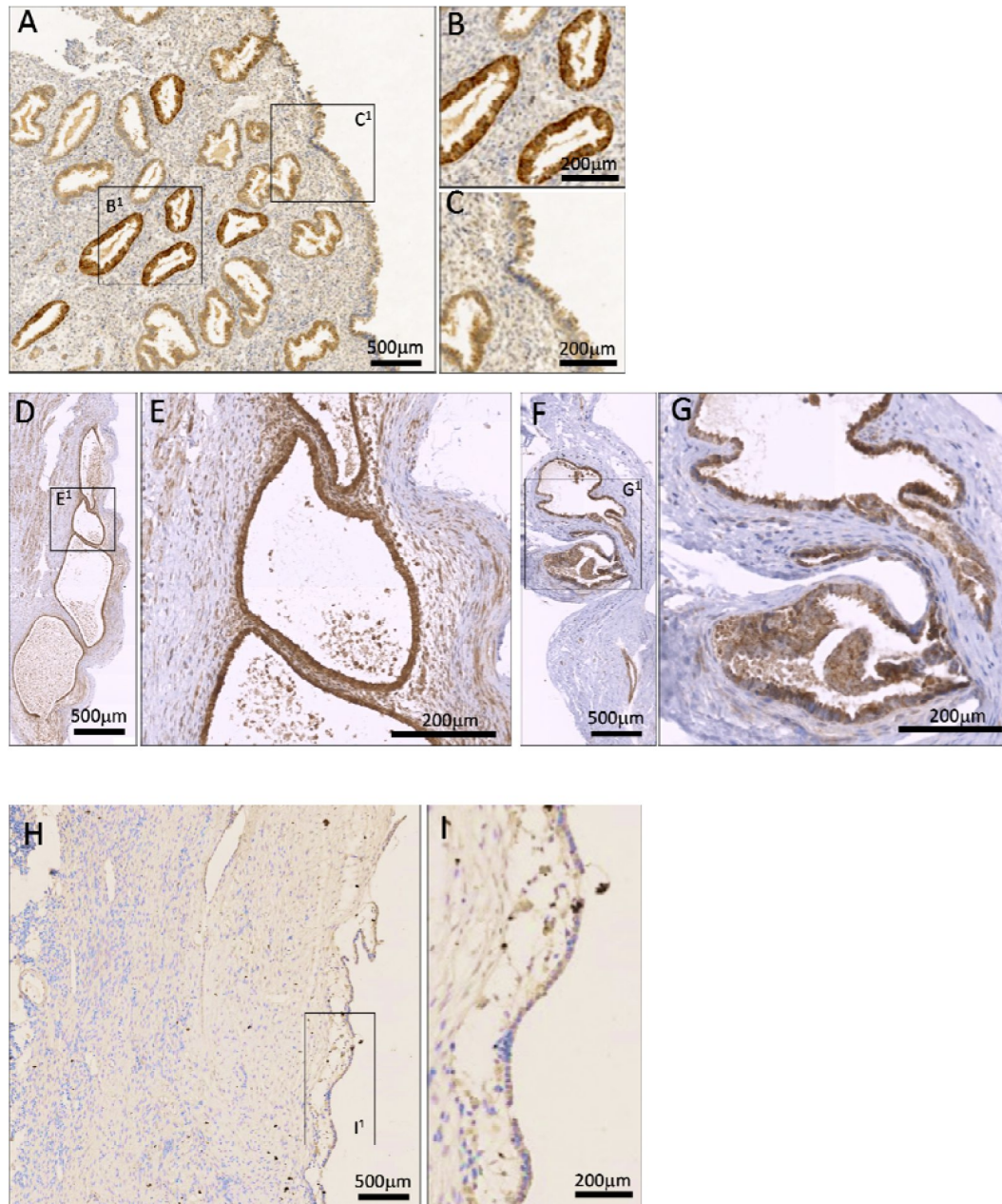


Figure 1.

Immunohistochemistry of KMO expression in human endometrium and distended endometriosis gland-like lesions.

Fixed tissue sections were stained with anti-KMO antibody (1:500 dilution) and visualized with DAB as described in the Methods section. **Panel A.** Normal human endometrium (200 X magnification); **B¹** and **C¹** insets denote areas shown in panels **B** and **C** at higher magnification. KMO expression is demonstrated as dark brown DAB-positive staining and is most intense in the epithelial cells lining the glands. **Panels D through G.** Human peritoneal endometriosis tissue lesions stained with anti-KMO antibody visualized with DAB. **D/E** PIN3652, Stage II, **F/G** PIN3306 Stage I. Note intense staining of cells lining glandular structures. **Panel H.** Ovarian-type endometriosis tissue lesion with higher magnification inset (**I¹**) showing KMO expression present but at lower intensity in the mesothelial tissue surface (this sample does not have epithelial cells in the lesion).

fold increase in KA at maximal inhibition was approximately 25-fold compared to the level seen after KNS898 0.01 mg/kg (**Fig. 2c**). KMO blockade resulted in a statistically-significant reduction of 3HK in plasma: KNS898 dosed at 0.01 mg/kg b.d. resulted in a mean ($\pm$ S.E.M.) plasma level of 3HK of 27.7 ± 7.2 ng/mL, 5 mg/kg resulted in 4.2 ± 0.3 ng/mL, and 25 mg/kg gave 0.9 ± 0.4 ng/mL. The difference between groups was statistically significant by one-way ANOVA with post hoc Tukey's test ($P = 0.001$). The fold decrease in 3HK at maximal inhibition was approximately 30-fold compared to the level seen after KNS898 0.01 mg/kg (**Fig. 2d**). Overall, there was a clear dose response to KNS898 administration leading to maximal KMO blockade at 25 mg/kg b.d. This dose was therefore selected for efficacy experiments going forward. A diagrammatic representation of the kynurenine pathway is shown as **Figure 2e** for reference.

KMO blockade reduces endometrial gland-like lesion burden in experimental endometriosis in mice

The experimental design for the mouse model of endometriosis is shown in **Figure 3a**. The pharmacological effect of KNS898 therapy showed appropriate levels of KNS898 detected in plasma (**Fig. 3b**), with accumulation of kynurenine (**Fig. 3c**), inhibition of 3HK production (**Fig. 3d**) and diverted metabolism of accumulated kynurenine to kynurenic acid (**Fig. 3e**). All recipient mice inoculated with donor tissue (groups G3, G4, and G5) developed distended endometrial gland-like structures (DEGLS). The incidence of DEGLS formation was enumerated at autopsy, and the axial length of each DEGLS was measured after excision from the surrounding tissue. In G3 (endometriosis + vehicle), 8 of 15 (53%) of the inoculated animals had developed DEGLS. In KNS898-treated group G4 (endometriosis + treatment from Day 19), DEGLS formed in 4 of 15 mice (26.7%) and in G5 (Endo + treatment start on Day 26) in 6 of 15 mice (40%). As expected, no DEGLS were formed in the non-inoculated control and sham groups. The total number of DEGLS per animal in each group was highest in G3 with an average of 4.0 per animal with DEGLS (total = 32 DEGLS in 8 mice in G3). Mice with endometriosis receiving KNS898 from the time of inoculation (G4) had an average of 2.0 DEGLS per animal with DEGLS (total = 8 DEGLS in 4 mice in G4) and those receiving KNS898 1 week after inoculation (G5) had an average of 1.8 DEGLS per animal (total = 11 DEGLS in 6 mice in G5) (**Figs. 3f** and **3g**). Statistical analysis by ANOVA showed a significant difference in endometriosis DEGLS burden between groups ($P = 0.0295$ for DEGLS per animal; $P = 0.004$ for DEGLS per group). DEGLS axial length and derived volume did not differ between groups (Supplementary Fig. 1a and b). All recipient mice inoculated with donor tissue lost body weight following inoculation which then gradually recovered. After recovery, body weight of all three inoculated groups was lower compared to the control groups for the duration of the study. Overall, there was no significant difference in body weight between G3 and the KNS898 treatment groups G4 and G5 (endometriosis + treatment from Day 26) (Suppl. Fig. 1c).

KMO is expressed in experimental endometriosis in mice

Histological examination of DEGLS identified them as containing cystic structures lined with epithelial layers identifiable as columnar epithelium, pseudostratified epithelium, squamous epithelium, and cuboidal epithelium, with goblet cells. These DEGLS were considered to represent endometriosis-like lesions derived from the implanted basal endometrial/myoepithelial layers of the donor mice uteri (Suppl. Fig 2). Immunohistochemistry using an antibody to KMO showed KMO protein expression localised mainly to the epithelial cells lining of the DEGLS, with a lesser degree of KMO positive staining in the closest surrounding connective tissue, in keeping with the previously observed KMO expression pattern in human endometriosis lesion tissue (**Fig. 4a** and **4d** and Supplementary Fig S3). The thickness (area divided by length) of the KMO positive epithelial layer was quantified for each DEGLS section using QuPath and there was no difference between groups G3, G4 and G5 (**Fig. 4b** and **4e**). However, quantification of KMO expression confirmed the high intensity of KMO staining in the epithelial lining layers (**Fig. 4c** and **4f**), but also showed a clear and statistically significant reduction in KMO expression intensity in those

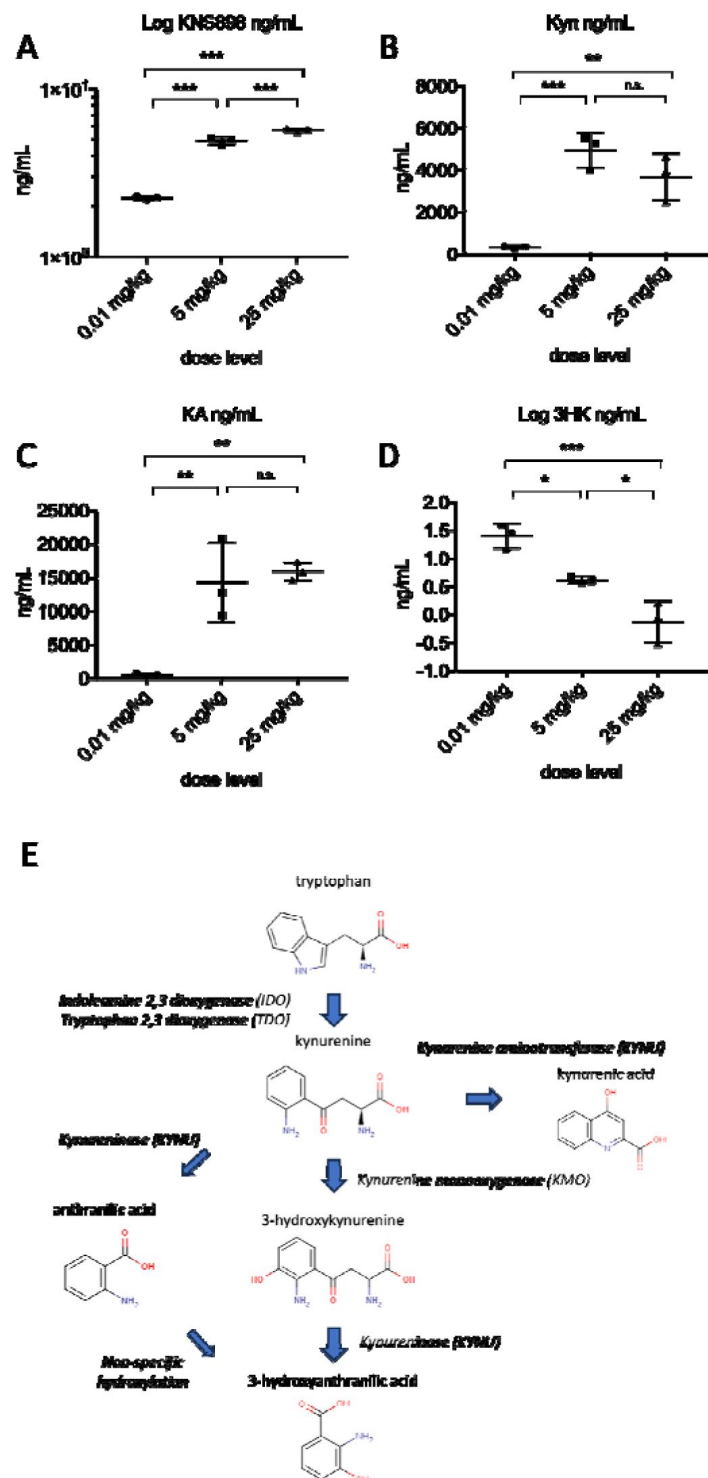


Figure 2.

KNS898 plasma levels and pharmacodynamic effect of KMO blockade.

Mice (n=3 per group, individual data shown) were given KNS898 twice daily by gavage at the doses shown. After 7 days, blood was sampled at euthanasia and KNS898 levels and kynurenine pathway metabolite levels measured by LC-MS/MS. **A.** KNS898 drug levels. **B.** Kynurenine. **C.** Kynurenic acid. **D.** 3-hydroxykynurenine (logarithmic scale). Comparison between groups by one way ANOVA with post hoc Tukey's test. *P < 0.05, **P < 0.01, ***P < 0.001, n.s. not statistically significant. **E.** A diagram of the kynurenine pathway showing the key step catalyzed by KMO.

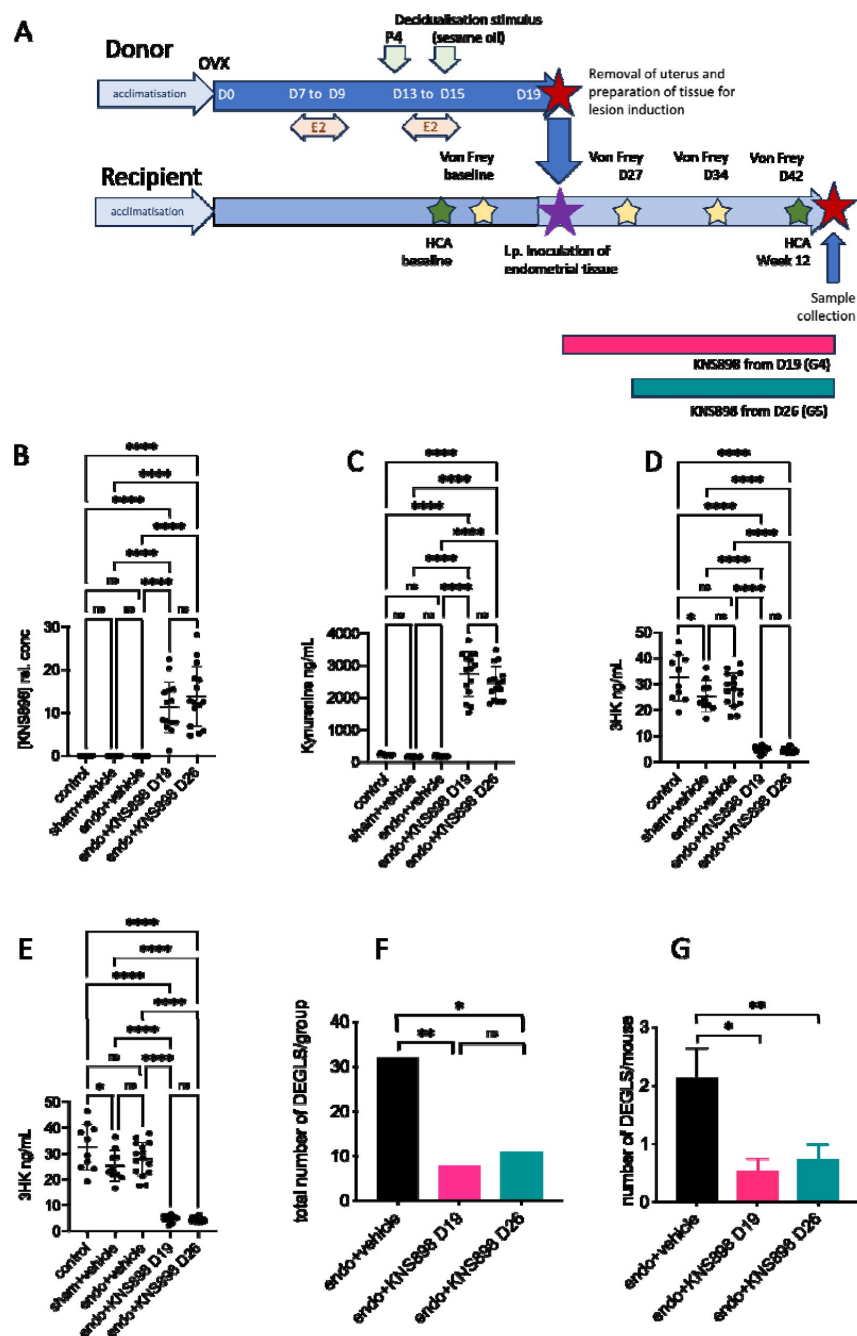


Figure 3.

Therapeutic effect of KNS898 in an experimental mouse model of endometriosis.

A. Experimental design. Ovariectomized (OVX) donor mice were hormonally stimulated as shown (E2 estradiol, P4 progesterone). At Day 19, donor mouse endometrial fragments were inoculated into recipient mice in a 1:1 ratio. KNS898 treatment 25 mg/kg twice daily by oral gavage was commenced at Day 19 or after a 1 week interval on Day 26 and in both cases continued for 2 weeks. Groups were G1: n=10, control mice; G2: n=10, sham-operated control mice; G3: n=15, endometriosis + vehicle; G4: n=15, endometriosis with KNS898 commenced at Day 19; G5: n=15, endometriosis with KNS898 commenced at Day 26. **B.** KNS898 drug levels (relative concentrations). **C.** Kynurenine. **D.** 3-hydroxykynurenine. **E.** Kynurenic acid. Individual data are shown in panels B through E. **Panel F and G.** Enumerated distended endometriosis gland-like structures (DEGLS) in recipient mice by treatment group. **F.** Total number of DEGLS per group. **G.** Total number of DEGLS per animal for all animals in the group; bars show mean with s.e.m. (**G**). Comparison between groups by one way ANOVA with post hoc Tukey's test. *P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001, n.s. not statistically significant.

areas in DEGLS removed from mice treated with the KMO inhibitor KNS898 (**Fig. 4g** [↗](#); $P = 0.008$). Representative micrographs from each of groups G3, G4 and G5 are presented in **Figures 4h, 4i** [↗](#) and **4j** [↗](#).

KMO inhibition reduces mechanical allodynia in experimental endometriosis

Clinical endometriosis is associated with visceral hyperalgesia and central sensitisation to pain¹¹ [↗](#). Visceral and central hyperalgesia may be tested in rodents using the Von Frey filament test¹² [↗](#). Baseline reaction values for hind paw and bladder Von Frey tests showed no significant difference in mechanical allodynia before inoculation. In established endometriosis without treatment (group G3), the mechanical allodynia threshold in the hind paw was statistically significantly lower compared to baseline for the group. When compared to the control groups at the corresponding time point beginning 1 week after inoculation and continuing until the end of the study, Day 26 KNS898-treated group (G5) showed a statistically-significant improvement in mechanical allodynia in the hind paw using the Von Frey test compared to mice in G3 with untreated endometriosis given vehicle control (Two-way ANOVA, Group effect $P = 0.003$, time effect $P < 0.0001$; Dunnett's multiple comparison test G5 vs G3 $P = 0.001$) (**Fig. 5a** [↗](#)). The mechanical allodynia threshold for the bladder reflex also was lower in mice with endometriosis compared to baseline throughout the study, and KNS898 treatment commencing at D26 (G5) was associated with a statistically significant improvement in bladder mechanical allodynia threshold at D42 compared to mice with untreated endometriosis given vehicle control (G3)(Two-way ANOVA, Group effect $P = 0.038$, time effect $P < 0.001$; Dunnett's multiple comparison test G5 vs G3 $P = 0.021$) (**Fig. 5b** [↗](#)).

KMO inhibition rescues impaired cage exploration behaviour and mobility in mice with endometriosis

HCA peripheral moving speed, time at cage edge, and illness behaviour, including temperature, motility and cage exploration behaviour was quantified using Home Cage Analysis (HCA). Baseline HCA was recorded before inoculation and at the end of the experiment. Mice with endometriosis without treatment showed an overall reduction in activity in moving distance and moving speed relative to baseline, and compared to sham-operated control groups, indicating a negative effect on behaviour due to endometriosis. Importantly, mice with endometriosis treated with KNS898 showed marked improvement in motility and cage exploration behaviour compared to untreated endometriosis mice, and although this difference between groups was statistically significant by Welch's one-way ANOVA, post hoc testing (Dunnett's T3) was not significant between groups. This qualitative difference of time spent exploring the periphery of the cage being lower for mice with endometriosis treated with vehicle control was seen in both the day and night phases (**Figs. 5c** [↗](#) and **5d** [↗](#)). A similar improvement with KNS898 treatment compared to vehicle control mice with endometriosis was seen for total moving distance, total moving speed, and peripheral distance, but not for total moving or climbing time, for which no difference between groups was detected. Together, these data indicate that KMO inhibition with KNS898 results in an improvement in well-being evidenced by improved cage exploration behaviour in addition to improved objective histological measures of endometriosis disease burden.

Discussion

In this study, we set out to investigate the potential for KMO inhibition as a non-hormonal therapy for endometriosis. First, we confirmed that KMO was expressed in human endometrium by immunohistochemistry, and then showed that KMO was clearly expressed in the epithelial cells in

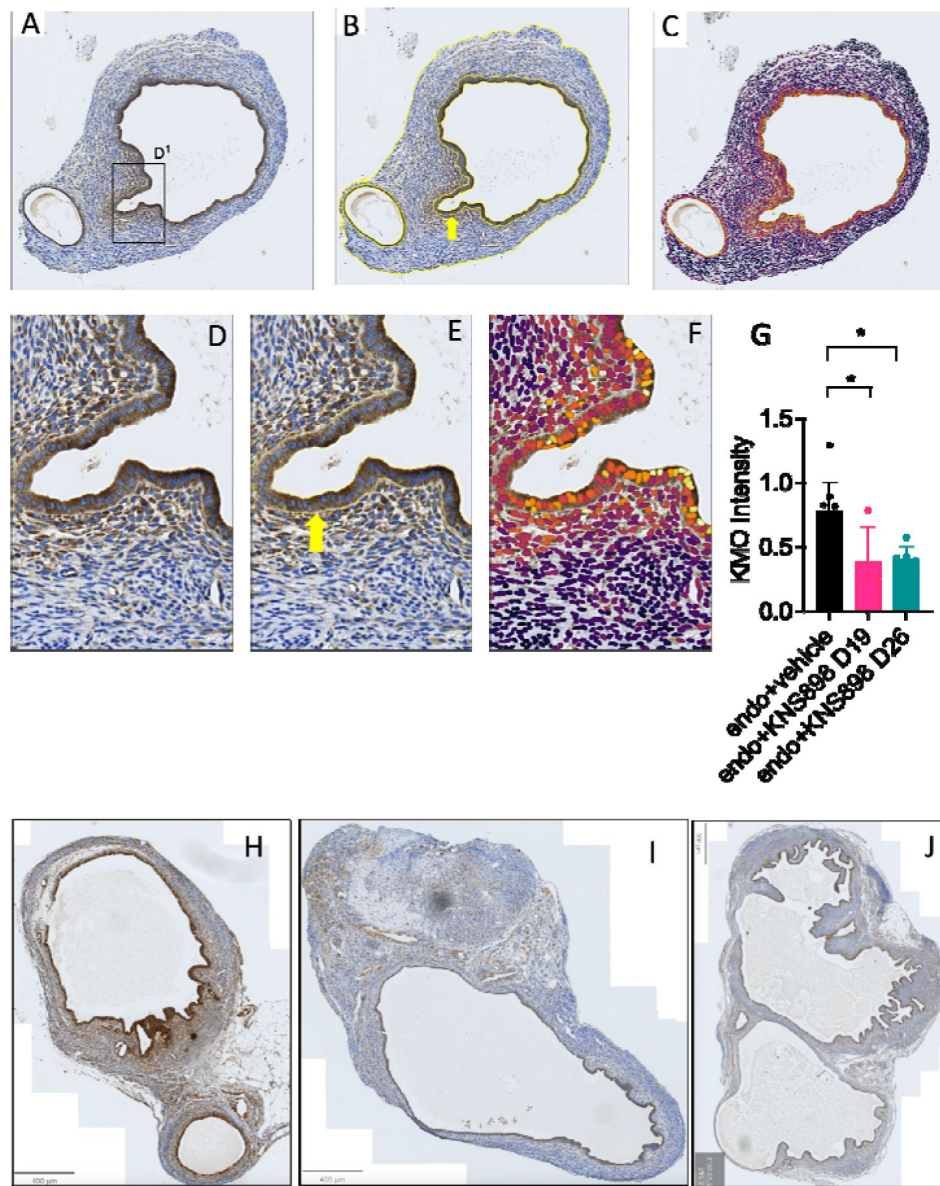


Figure 4.

Quantification of KMO expression in mouse model distended endometriosis gland-like structure (DEGLS) lesions.

Sections were visualized at 200 X magnification (A, B, C) with higher magnification insets shown (D, E, F). Panel A and D. Fixed tissue sections were stained with anti-KMO antibody (1:500 dilution) and visualized with DAB as described in the Methods section. B and E. QuPath was used to identify the epithelial tissue layers (yellow arrows denote the boundary) which were quantified by thickness. C and F. KMO expression intensity quantified and heat map expression values are overlaid. G. KMO expression staining intensity per unit area of endometriosis DEGLS epithelium, categorized by treatment group (G3 endometriosis + vehicle; G4 endometriosis + KNS898 from D19; G5 endometriosis + KNS898 from D26). Individual data points shown. Comparison between groups by one way ANOVA with post hoc Tukey's test. *P < 0.05. H, I and J. Representative micrographs from G3 (H), G4 (I) and G5 (J) showing KMO expression in the epithelial cells lining each DEGLS.

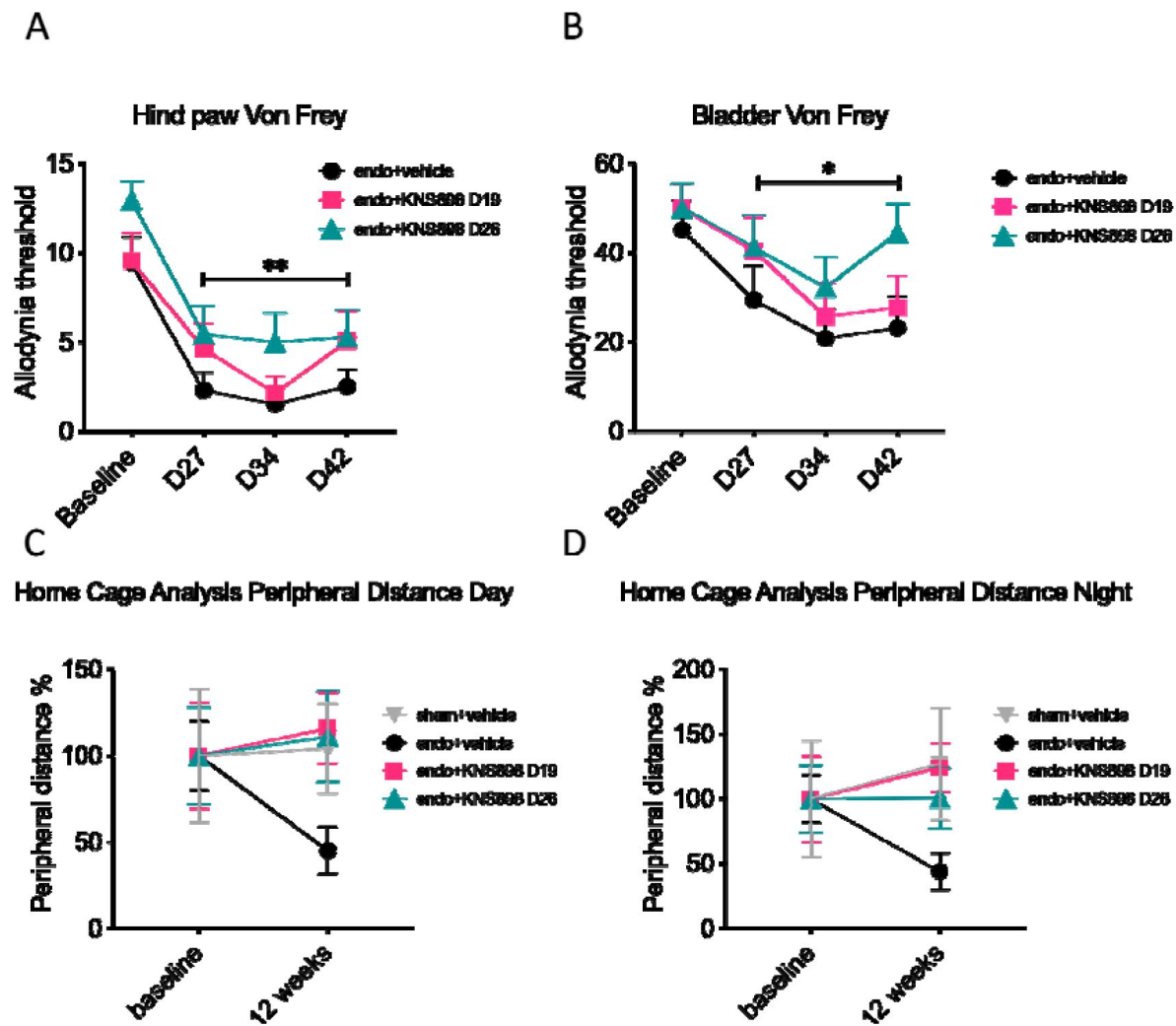


Figure 5.

Effect of KNS898 on mechanical allodynia and illness behaviour in an experimental mouse model of endometriosis.

A. Hind paw Von Frey filament test showing effect of endometriosis and KNS898 treatment in groups G3, G4 and G5 **B.** Bladder Von Frey filament test **C.** Home Cage Analysis of motility showing a daytime motility deficit in mice with endometriosis compared to control mice, and clear restitution of normal motility in KNS898 treated groups. **D.** Nighttime home cage motility analysis showing the benefit of KNS898 treatment on normalizing the motility deficit seen in mice with endometriosis. Data are mean with s.e.m. For A and B, statistical comparison between groups was by two-way ANOVA to compare Group effect and Time effect, with multiple group comparison using Dunnett's T3 test. Asterisks represent treatment group effect statistical significance of the Dunnett's T3 test comparing treatment group G5 to G3 vehicle control * $P < 0.05$, ** $P < 0.01$. For C and D, Welch's ANOVA with multiple group comparison using Dunnett's T3 test was used. Although the ANOVA was statistically significant, the post hoc Dunnett's T3 was not, therefore no asterisks are marked.

human endometriosis lesions. Next, we demonstrated that the highly specific KMO inhibitor KNS898 was orally bioavailable when given twice daily by gavage in mice, and clearly blocked KMO activity in a dose-dependent manner at a dose of 25mg/kg. We therefore used that dose to test the efficacy of KMO blockade with KNS898 in mice with experimentally-induced endometriosis. One important finding of this project is that KMO blockade resulted in a reduction in endometriosis severity compared to untreated mice with endometriosis, specifically in terms of reducing i) the number of mice that developed endometriosis tissue lesions, and ii) the number of lesions per mouse in those that did develop lesions. The histopathology of the experimental endometriosis lesions was sufficiently similar macroscopically to that seen in the examined human DEGLS, and KMO was evidently highly expressed in the same tissue distribution in model lesions compared to human disease. KMO blockade also decreased lesion KMO expression. Critically, and importantly from a translational perspective, therapeutic blockade of KMO improved visceral hyperalgesia measured by reduced mechanical allodynia and restored normal cage exploration behaviour and mobility in treated mice compared to untreated mice with endometriosis. Together, these data show that KMO is expressed in human and mouse endometriosis tissue lesions and that therapeutic KMO blockade reduces the number of endometriosis lesions and improves holistic metrics of disease behaviour in mice.

The model of endometriosis, using inoculation of endometrial tissue of ovariectomized donor mice, reliably induced the pathophysiological symptoms indicative of endometriosis in recipient mice. Test groups inoculated with endometrial tissue (G3-G5) showed significant growth of ectopic endometrial tissue. Groups treated with test article experienced significantly less DEGLS development (significantly fewer DEGLS were noted in treated groups when compared to vehicle treated groups). Disease burden in the treatment group that had treatment starting immediately after inoculation was lower than that of the vehicle-only treated group. It is not clear why mean cystic size and cystic volume in treated animals was not smaller in treated animals. We can only speculate that KMO blockade may potentiate rapid involution of cysts, but this cannot be proven mechanistically here.

Mice that received inoculated endometrial tissue showed a measurable and increased visceral hyperalgesic pain response (lower mechanical threshold) as measured by bladder response to von Frey filament testing, and improvement in a surrogate marker of central sensitisation to pain measured by hind paw Von Frey filament testing when compared to control mice. The mechanical threshold of both treatment groups trended higher compared to the vehicle treated group. One interpretation of these data is that KMO inhibition reduced responses to pain caused by the presence of endometriosis, i.e. improving visceral hyperalgesia.

Home cage behaviour using HCA indicated a reduced overall activity in endo-inoculated mice when compared to control mice. It should also be noted that test article-treated animals in both groups showed more locomotor behaviour and a seemingly better quality of life within the home cage environment when the home-cage dynamics were monitored. This supports the notion that treated mice exhibit less propensity for behaviours that are, at times, typical of depressive and anxiety-like behaviour in home-cage, group housed conditions.

Under the experimental conditions imposed, treatment with KNS898 on two dosing schedules provided a significant reduction in DEGLS formation within the inoculated mice, as well as a seemingly higher pain threshold. This attenuation of pain response was coupled with increased activity levels in the home cage. Taken together, these data suggest a therapeutic effect to alleviation of certain salient symptoms of endometriosis, as well as a reduction in the number and size of DEGLS.

Non-pathological endometrium is a site of high KMO expression. Because endometriosis lesions in women are 'endometrial-like' tissue rather than normal endometrium, we tested, and demonstrated expression of KMO in the epithelial layers of endometriosis lesions sampled from

women undergoing surgery for endometriosis. First-line medical treatment for endometriosis is the contraceptive pill or other ovarian steroid hormone suppressive drugs. Treatment failures are frequent, side effects are common, all approaches are contraceptive. Many women opt for invasive surgery to remove or ablate the endometriosis lesions.

In conclusion, KMO is expressed in human endometriosis tissue lesions and in a mouse model of endometriosis in the epithelial layers of distended endometrial gland-like structures. Oral KNS898 reliably induced a biochemical state of KMO blockade with accumulation of kynurenine, diversion to kynurenic acid and ablation of 3-hydroxykynurenine production. KMO blockade improved histological and symptomatic behavioural endometriosis disease features with an overall benefit, even when treatment commenced one week after establishment of the disease. KMO blockade is therefore a promising avenue for a new non-hormonal therapeutic modality for endometriosis.

Materials and methods

Ethical approvals and permissions

The human tissue samples were obtained from participants who had given fully informed written consent under ethical approval granted by Lothian Research Ethics Committee (LREC 11/AL/0376). Human tissue samples were obtained with ethical approval and fully informed consent from individuals attending the Royal Infirmary of Edinburgh as described below. Animal experiments conducted by NAASON Inc were carried out according to the National Institute of Health (NIH) & National Institutes of Health Korea (NIHK) guidelines for the care and use of laboratory animals and approved by Naason Science in accordance with all applicable FELASA, IACUC and AAALAC guidelines. Animal experiments outsourced to Syneos Health were conducted with institutional ethical approval.

Human Patients and Samples

Tissue samples were collected from patients undergoing a diagnostic laparoscopy for suspected endometriosis following Endometriosis Phenome and Biobanking Harmonisation Project (EPHect) guidelines¹³. Patient summary characteristics are presented in Supplementary Table S1. Note there was a range of disease stages assigned at time of surgery according to American Fertility Society (AFS) criteria¹⁴. Cycle stage was determined by measuring hormones in blood according to standard protocols and assessment of eutopic endometrial tissue histology when such samples were available¹⁵. Lesions were recovered from 17 patients, of these n=10 were recovered from the peritoneal side wall consistent with classification as superficial peritoneal endometriosis lesions and n=5 from the cysts of ovarian disease (endometrioma) (n=2 on hormones). Eutopic endometrium was from 4 patients n=3 of which had no lesions at time of surgery (noted as stage 0). General histology of samples was assessed using H&E staining.

Immunohistochemistry (human endometrium and human endometriosis tissue lesions)

5 µm sections of formalin-fixed paraffin-embedded tissue blocks were mounted on SuperFrost Plus adhesion slides (Thermo Fisher Scientific). Sections were deparaffined with xylene and rehydrated prior to heat-induced antigen retrieval using Instant Pot: Tris-EDTA pH9¹⁶. Sections were washed with tap water and incubated in phosphate buffered saline (PBS) for 5 minutes. Endogenous peroxidase was blocked with 0.3% hydrogen peroxide in 70% v/v methanol for 30 mins at room temperature then washed in PBS prior to blocking in Normal Goat Serum (NGS)/PBS/bovine serum albumin (BSA)(5%) for 30 mins and streptavidin for 15 mins. Sections were washed twice in PBS and then blocked with biotin for 15 mins and washed in PBS. The primary antibody to KMO (KMO Rabbit polyclonal, Proteintech, Catalog Number:10698-1-AP)¹⁷ was diluted to a final concentration of 1:1000 in NGS/PBS/BSA and incubated overnight at 4°C in a

humidity chamber. Sections were washed twice with PBS/Tween 0.05% (1ml Tween in 2L PBS) for 5 mins. The secondary detection antibody Goat Anti-Rabbit Biotinylated (Vector Cat number: BA-1000) was diluted in NGS/PBS/BSA (1:500) and incubated for 30 mins, prior to washing twice in PBS/Tween 0.05%, for 5 mins before adding the detection system reagent Streptavidin-HRP (DAKO Cat Number P0397) 1:500 in PBS for 30 min, washed and stained with DAB (DAKO Cat Number K3468) as per manufacturer's directions and incubated for 5 mins before a final wash with tap water. Sections were counterstained with Haematoxylin, dehydrated through graded ethanol and mounted. Sections of stained slides were scanned on a Zeiss Axioscan Z1 slide scanner and exported as TIFF files: images were evaluated for stromal, epithelial and immune cell content.

KNS898 preparation for oral administration

KNS898 powder was weighed and dissolved at the required concentrations in a final vehicle of 2% DMSO, 20% PEG200, 78% 0.15M NaCl by volume. Brief sonication on ice was done to facilitate dissolution.

In vivo confirmation of KMO inhibition by KNS898 in mice

This experiment was outsourced to Syneos Health (Les Templiers, 2400 route des Colles, 06410 Biot, Sophia-Antipolis, France). A formal pharmacokinetic/pharmacodynamic study was not required at this stage. Female C5Bl/6J mice aged 12 weeks were purchased from Charles River Laboratories, maintained on standard 12 hour light-dark cycle, given free access to water and standard chow before being randomised to one of three dose levels of KNS898 (n=3 per group, total n=9 mice). Dose levels tested were 0.01mg/kg, 5mg/kg and 25mg/kg. Mice were gavaged with 0.5 mL of drug in vehicle twice daily for 7 days before euthanasia and plasma sampling.

Plasma samples

Blood was sampled into Sarstedt Microvette CB K2EDTA 300 μ L tubes and centrifuged at 5,000 rpm (2380 RCF) for 3 mins. Plasma was aliquoted, frozen on dry ice and transferred to storage at -80°C prior to temperature-controlled shipping.

LC-MS/MS analysis of plasma drug levels and kynurenine metabolites

Plasma samples (100 μ L) were diluted at a 1:1 ratio with 4% phosphoric acid and enriched with 50 ng $^{13}\text{C}_6$ -kynurenine, $^{13}\text{C}_6$ -3-hydroxykynurenine (Sigma Aldrich, custom synthesis) and d5-kynurenic acid (CDN isotopes). 12-point calibration standards (0.1 to 100 ng) were prepared for KNS898, kynurenine (KYN), kynurenic acid (KA), and 3-hydroxykynurenine (3HK) and extracted alongside samples using solid phase extraction plates (Waters Oasis HLB, 10 mg sorbent, 30 μ m particle size). Extracts were dried down under nitrogen and reconstituted in LC-MS grade water (100 μ L). 10 μ L was injected onto a column (Ace C18-PFP column; 100 x 2.1 mm internal diameter 1.7 μ m; HiChrom (VWR, Lutterworth)) using an Acquity I-Class UPLC liquid chromatography system (Waters) linked to a QTRAP 6500+ mass spectrometer (AB Sciex).¹⁰ The flow rate was set at 0.4 mL/min with a column temperature of 40°C . Separation was carried out using a gradient mobile phase system of A – 0.1% aqueous formic acid and B – 0.1% formic acid in methanol, starting at 15%B, rising to 85%B over 6 mins and returning to 15%B by 9 mins. Mass spectrometry settings were for positive mode electrospray (5.5 kV, 700°C) and multiple reaction monitoring m/z 209.0 $\rightarrow$ 192.2 for KYN, m/z 189.9 $\rightarrow$ 144.1 for KA m/z 225.0 $\rightarrow$ 208.0 for 3HK and m/z 361.1 $\rightarrow$ 120.1 for KNS898 and for internal standards were m/z 231.0 $\rightarrow$ 214.0 for $^{13}\text{C}_6$ -3HK, m/z 195.1 $\rightarrow$ 177.2 for d5-KA and m/z 215.0 $\rightarrow$ 197.8 for $^{13}\text{C}_6$ -kynurenine. Retention times for KYN, KA, 3HK and KNS898 were 1.8, 3.2, 1.2 and 6.5 mins, respectively and 3.2 mins for d5KA, 1.8 mins for $^{13}\text{C}_6$ -3HK and 1.2 mins for $^{13}\text{C}_6$ -KYN. Data were acquired by Analyst 1.7.1 software (AB Sciex) and linear regression

analysis was carried out on MultiQuant 3.0.3 software (AB Sciex) where peak integrations and amounts of each kynurenine metabolite and KNS898 were calculated using the peak area ratio of compound/internal standard, with data further handled in Microsoft Excel 2016 as described.¹⁸

Experimental mouse model of endometriosis

This experiment was outsourced to Naason Science Inc., Osong, Korea (KBIO New Drug Development Center #506, Chungbuk, Korea, 28160) using protocols originally developed by the Saunders team in Edinburgh.^{19,20} Experimental design and groups are shown in **Figure 3a** and Supplementary Table S2. There were 5 groups of mice with $n=10-15/\text{group}$: 25 mg/kg KNS898 was administered twice a day via oral gavage in two of the groups of mice. Group 4 received KNS898 from the time of endometrial tissue inoculation (Day 19; G4); group 5 commenced dosing 1 week after inoculation (Day 26; G5). Group 3 received vehicle (2% DMSO, 20% PEG200, 78% 0.15M NaCl) in the same regimen.

To perform the mouse model of endometriosis, donor female C57Bl/6 mice aged 6 weeks were acclimatized for 2 weeks prior to surgery. Ovariectomy (Day 0) was performed at 8 weeks of age under general anaesthesia with monitoring, with analgesia that extended to the post-operative period with buprenorphine (0.03 ml) (Veterges ic® 3 mg/ml, Ceva Inc., Korea) subcutaneously. To prepare donor tissue that would best replicate menstrual-like tissue in women, ovariectomized (OVX) mice were primed with daily s.c. injections of 100 ng 17 β -estradiol (E2) on days 7, 8 and 9. On days 13 – 19 a silastic progesterone (P4) pellet was implanted subcutaneously. These animals were injected once daily with E2 (5 ng in sesame oil) on days 13, 14 and 15. Decidualization was induced in one uterine horn with an injection of 20 μ l sesame oil 4 hours after the last E2 injection. On day 19 (4 days after induction of decidual response), donor mice were killed 4 hours after removal of the P4 pellet. Endometrial tissue was then scraped from the myometrial layer of the decidualized uterine horn, suspended in 500 μ l PBS and injected via a XG needle sprayed into the lower abdominal cavity of the recipient mouse under general anaesthesia with monitoring and post-operative analgesia as described. The ratio of donor to recipient mouse was 1:1 (from one donor to one recipient). Recipient mice had intact ovaries to ensure ongoing hormonal stimulation of the injected tissue: group allocations are shown in Supplementary Table S2.

Mechanical allodynia test by the Von Frey method

Abdominal and hind paw Von Frey tests were performed in the recipient animals before inoculation (baseline), and 1, 2, and 3 weeks after inoculation. Mechanical threshold was measured using Von Frey filaments. For the hind-paw, 15, 8, 6, 4, 2, 1.4, 0.6, 0.4 g filaments were used, and for the bladder reflex to filament application to the lower abdomen, 60, 26, 10, 8, 6, 4, 2, 1 g filaments were used. The experimenter was blind to the group allocation in order to reduce bias.

Cage exploration and behavioural assay using Home Cage Analysis

Home Cage Analysis (HCA) was performed in the recipient animals before inoculation (baseline), and at the late stage of treatment. Recipient mice were randomly housed using a Monte Carlo randomization. All animals had a micro-chip (BioMark, USA) inserted to the abdomen prior to being placed in the home-cage. This procedure does not cause undue discomfort or hamper, in any way, animal movement. Total moving distance, total moving time, moving speed, isolation/separation distance, isolated time, peripheral time, peripheral distance, in centre zones time, in centre zones distance, climbing time, and body temperature were tracked automatically by an ActualHCA™ Home Cage Analyzer (ActualAnalytics Ltd., Edinburgh, UK) and processed with proprietary machine learning and artificial intelligence algorithms.

Endpoint tissue and plasma sampling

On experimental Day 40, all recipient mice were euthanized, and blood collected via cardiac puncture. Whole blood was collected into heparinised tubes, and plasma was separated by centrifugation (3000 rpm for 15 min) at 4°C. Separated plasma was collected in Eppendorf microtubes, frozen on dry ice and stored at -80°C. Photographs of the abdominal cavity were obtained. Lesions from the abdominal cavity were harvested. DEGLS were dissected from the surrounding abdominal tissue and measured for size and volume. DEGLS volume was measured using the following formula²¹; $\text{Volume} = \text{long diameter} \times (\text{short diameter}/2)^2 \times \pi$. DEGLS were fixed in a 4% paraformaldehyde solution and prepared for standard H&E. If more than one DEGLS was present in an animal, DEGLS that were not used for H&E were snap-frozen in liquid nitrogen and stored at -80°C.

Endometriosis lesion histology

Endometriosis lesion tissue blocks were sectioned at a uniform thickness of 5 µm and were mounted onto a microscope slide. The slide then underwent deparaffination and hydration. Paraffin was removed from the slide using xylene, then hydration through graded ethanol and washing were performed. Slides were then stained with Harris haematoxylin and alcoholic eosin Y and mounted after dehydration and clearing with xylene. The H&E-stained images were visualized using a Slide Scanner (Panoramic scan, 3D HISTECH).

Immunohistochemistry (mouse DEGLS)

Immunohistochemistry to detect KMO in mouse DEGLS tissue was performed on a Leica Bond III automated immunostaining robot. 5 µm thick sections obtained from FFPE (formalin Fixed Paraffin Embedded) samples mounted on superfrost plus slides were stained as follows. Heat induced epitope retrieval (HIER) was performed using Epitope Retrieval Solution 1 (Leica, ER1 pH 6.0 citrate based solution) for 20 minutes at 99°C. Tissue sections were then incubated for 10 minutes in hydrogen peroxide to block endogenous hydrogen peroxidase activity followed by 10 minute blocking with normal goat serum. The primary antibody against KMO (Proteintech 10698-1-AP @1:1000 Rabbit)¹⁷ was incubated for 1 hour, then incubated with a goat anti-rabbit peroxidase conjugated secondary antibody for 30 minutes prior to visualisation with diaminobenzoate (DAB) using standard protocols.

Digital slide scanning

Whole sections were scanned using a Zeiss Axioscan Z1 whole slide scanner. The image files (.czi) were batch converted to Tif format for image analysis and quantification, acquisition and batch export used Carl Zeiss Zen v2.5 software.

Quantitative image analysis

Image analysis was carried out using QuPath v0.4.2. Analysis was standardised by using multiple regions of interest from individual slides and collating them into a training image. To improve stain contrast, overlapping DAB and haematoxylin staining was deconvoluted by manual optimisation of the stain vectors. Individual slide epithelia were annotated, with relative DAB optical densities and annotation shape measurements taken. Epithelial thickness and area were calculated to enable correlation with KMO intensity. To generate the heat map of cell KMO expression, a cell detection was carried out on haematoxylin staining using standard parameters, a 5-100µm² area range, cell expansion of 1µm and a threshold of 0.14.

Statistical analysis

Power calculations were performed using G*Power (v3.1.9.4) software. Input parameters were used: 2-tailed, α -error probability = 0.05, and power ($1-\beta$ error probability) = 0.80. Continuous variable data were tested for Normality of distribution with a one sample Kolmogorov-Smirnov test. Normally distributed data were analysed by one-way ANOVA with post-hoc Dunnett's T3 for multiple groups. Data comparing treatment group effects at multiple time-points were analysed by two-way ANOVA with multiple comparison testing by Dunnett's method. Data comparing multiple groups with unequal variances were analyzed with Welch's ANOVA. Data not following the Normal distribution were analysed with non-parametric Kruskal-Wallis test. Categorical and proportions data were analysed by Fisher's exact test. Data were visualised with GraphPad Prism.

Acknowledgements

We would like to thank the University of Edinburgh MRC Confidence in Concept award team: Andrew McBride, Lorraine Jackson. We thank Susan Bodie, Dave Pritchard from Edinburgh Innovations. We thank all staff and support team members at Syneos Health and NAASON Science Inc.

Additional information

Funding

UKRI Medical Research Council Confidence-in-Concept grant MRC/CIC8/73 (DJM, SPW, PTKS, AH)
UKRI Medical Research Council Senior Clinical Fellowship MR/P008887/1 (DJM)

Author contributions

Conceptualization: DJM, SPW, PTKS, AH

Methodology: BH, IS, NZMH, SGD, JPS, MM, LB, LCP, PJS, AT, SPW, AH, PTKS, DJM

Investigation: BH, IS, XZ, FC, SGP, JPS, LB, HYL, YGK, KHP, LCP GF, AT, DC, TA

Visualization: BH, IS, KHP, LCP, DJM

Funding acquisition: DJM, SPW, PTKS, AH.

Project administration: DJM, XZ, LCP, PES, AT, PTKS, AH, NZMH

Supervision: MM, LCP, AT, PTKS, DJM

Writing – original draft: DJM, LCP

Writing – review & editing: BH, IS, XZ, FC, NZMH, SGD, JPS, MM, LB, LCP, PJS, AT, SPW, AH, PTKS, DJM.

Declaration of interests

The following authors have interests to declare: S.P.W., D.J.M. are co-founders of Kynos Therapeutics Ltd.. D.J.M. is a Board Member of Kynos. The University of Edinburgh controls Patents WO2015/091647, WO2016/097144, WO2016/188827 that relate to inhibitors of KMO

inhibitors, and include the compound used in this paper. The remaining authors declare no competing interests.

Data and materials availability

All data are available in the main text or the supplementary materials.

KNS898 availability is restricted under a Material Transfer Agreement. Please contact the corresponding author in the first instance.

References

1. Saunders P.T.K., Horne A.W. (2021) **Endometriosis: Etiology, pathobiology, and therapeutic prospects** *Cell* **184**:2807–2824 <https://doi.org/10.1016/j.cell.2021.04.041>
2. Dorning A., Dhami P., Panir K., Hogg C., Park E., Ferguson G.D., Hargrove D., Karras J., Horne A.W., Greaves E. (2021) **Bioluminescent imaging in induced mouse models of endometriosis reveals differences in four model variations** *Dis Model Mech* **14** <https://doi.org/10.1242/dmm.049070>
3. Horne A.W., Saunders P.T.K., Abokhras I.M., Hogg L. (2017) **Top ten endometriosis research priorities in the UK and Ireland** *Lancet* **389**:2191–2192 [https://doi.org/10.1016/S0140-6736\(17\)31344-2](https://doi.org/10.1016/S0140-6736(17)31344-2)
4. Mole D.J. *et al.* (2016) **Kynurenine-3-monooxygenase inhibition prevents multiple organ failure in rodent models of acute pancreatitis** *Nat Med* **22**:202–209 <https://doi.org/10.1038/nm.4020>
5. Thul P.J., Lindskog C. (2018) **The human protein atlas: A spatial map of the human proteome** *Protein Sci* **27**:233–244 <https://doi.org/10.1002/pro.3307>
6. Atkins H.M., Bharadwaj M.S., O'Brien Cox A., Furdul C.M., Appt S.E., Caudell D.L. (2019) **Endometrium and endometriosis tissue mitochondrial energy metabolism in a nonhuman primate model** *Reprod Biol Endocrinol* **17** <https://doi.org/10.1186/s12958-019-0513-8>
7. Liu X.T. *et al.* (2018) **Indoleamine 2,3-dioxygenase suppresses the cytotoxicity of 1 NK cells in response to ectopic endometrial stromal cells in endometriosis** *Reproduction* **156**:397–404 <https://doi.org/10.1530/REP-18-0112>
8. Liddle J. *et al.* (2017) **The discovery of potent and selective kynurenine 3-monooxygenase inhibitors for the treatment of acute pancreatitis** *Bioorg Med Chem Lett* **27**:2023–2028 <https://doi.org/10.1016/j.bmcl.2017.02.078>
9. Walker A.L. *et al.* (2017) **Development of a Series of Kynurenine 3-Monooxygenase Inhibitors Leading to a Clinical Candidate for the Treatment of Acute Pancreatitis** *J Med Chem* **60**:3383–3404 <https://doi.org/10.1021/acs.jmedchem.7b00055>
10. Hayes A.J. *et al.* (2023) **Kynurenine monooxygenase regulates inflammation during critical illness and recovery in experimental acute pancreatitis** *Cell Rep* **42** <https://doi.org/10.1016/j.celrep.2023.112763>
11. Ryan A., Healey M., Cheng C., Dior U., Reddington C. (2022) **Central sensitisation in pelvic pain: A cohort study** *Aust N Z J Obstet Gynaecol* **62**:868–874 <https://doi.org/10.1111/ajo.13596>
12. Deuis J.R., Dvorakova L.S., Vetter I. (2017) **Methods Used to Evaluate Pain Behaviors in Rodents** *Front Mol Neurosci* **10** <https://doi.org/10.3389/fnmol.2017.00284>

13. Rahmioglu N. *et al.* (2014) **World Endometriosis Research Foundation Endometriosis Phenome and Biobanking Harmonization Project: III. Fluid biospecimen collection, processing, and storage in endometriosis research** *Fertil Steril* **102**:1233–1243 <https://doi.org/10.1016/j.fertnstert.2014.07.1208>
14. American Society for Reproductive Medicine (1997) **Revised American Society for Reproductive Medicine classification of endometriosis: 1996** *Fertil Steril* **67**:817–821 [https://doi.org/10.1016/s0015-0282\(97\)81391-x](https://doi.org/10.1016/s0015-0282(97)81391-x)
15. Noyes R.W., Hertig A.T., Rock J. (1975) **Dating the endometrial biopsy** *Am J Obstet Gynecol* **122**:262–263 [https://doi.org/10.1016/s0002-9378\(16\)33500-1](https://doi.org/10.1016/s0002-9378(16)33500-1)
16. Kearns N., Norville K., Frelinger J.G. (2023) **Instant Pot for antigen retrieval: a simple, safe and economical method for use in immunohistochemistry** *Biotechniques* **74**:237–241 <https://doi.org/10.2144/btn-2022-0043>
17. Lai M.H., Liao C.H., Tsai N.M., Chang K.F., Liu C.C., Chiu Y.H., Huang K.C., Lin C.S. (2021) **Surface Expression of Kynurenine 3-Monooxygenase Promotes Proliferation and Metastasis in Triple-Negative Breast Cancers** *Cancer Control* **28** <https://doi.org/10.1177/10732748211009245>
18. Homer N.Z.M. (2023) **Using MultiQuant and Excel software to evaluate and report multi-analyte targeted LC-MS/MS data** *protocols.io*
19. Greaves E., Cousins F.L., Murray A., Esnal-Zufiaurre A., Fassbender A., Horne A.W., Saunders P.T. (2014) **A novel mouse model of endometriosis mimics human phenotype and reveals insights into the inflammatory contribution of shed endometrium** *Am J Pathol* **184**:1930–1939 <https://doi.org/10.1016/j.ajpath.2014.03.011>
20. Horne A.W., Ahmad S.F., Carter R., Simitsidellis I., Greaves E., Hogg C., Morton N.M., Saunders P.T.K. (2019) **Repurposing dichloroacetate for the treatment of women with endometriosis** *Proc Natl Acad Sci U S A* **116**:25389–25391 <https://doi.org/10.1073/pnas.1916144116>
21. Yun C.W., Yun S., Lee J.H., Han Y.S., Yoon Y.M., An D., Lee S.H. (2016) **Silencing Prion Protein in HT29 Human Colorectal Cancer Cells Enhances Anticancer Response to Fucoïdan** *Anticancer Res* **36**:4449–4458 <https://doi.org/10.21873/anticancer.10989>

Author information

Ben Higgins

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK

Ioannis Simitsidellis

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, EXPPECT Edinburgh, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, MRC Centre for Reproductive Health, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK

Xiaozhong Zheng

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK

Frances Collins

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, EXPPECT Edinburgh, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, MRC Centre for Reproductive Health, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK

Natalie ZM Homer

Mass Spectrometry Core, Edinburgh Clinical Research Facility, The University of Edinburgh, Edinburgh, UK, Centre for Cardiovascular Science, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK

Scott G Denham

Mass Spectrometry Core, Edinburgh Clinical Research Facility, The University of Edinburgh, Edinburgh, UK

Joanna P Simpson

Mass Spectrometry Core, Edinburgh Clinical Research Facility, The University of Edinburgh, Edinburgh, UK

Mike Millar

SuRF Molecular Histology Facility, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK

Lyndsey Boswell

SuRF Molecular Histology Facility, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK

Hee Y Lee

Naason Science, Inc., Cheongju, Republic of Korea

Yeon G Kim

Naason Science, Inc., Cheongju, Republic of Korea

Kyung H Park

Naason Science, Inc., Cheongju, Republic of Korea

Larry C Park

Naason Science, Inc., Cheongju, Republic of Korea

Patrick J Sweeney

Naason Science, Inc., Cheongju, Republic of Korea

Gerard Feraille

Syneos Health France, Biot, France

Alessandro Taddei

Syneos Health France, Biot, France

David Chagras

Syneos Health France, Biot, France

Thierry Alvarez

Syneos Health France, Biot, France

Scott P Webster

Centre for Cardiovascular Science, Queen's Medical Research Institute, The University of Edinburgh, Edinburgh, UK

Andrew Horne[†]

EXPPECT Edinburgh, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, MRC Centre for Reproductive Health, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK

[†]Senior authors

Philippa TK Saunders[†]

EXPPECT Edinburgh, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, MRC Centre for Reproductive Health, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK
ORCID iD: [0000-0001-9051-9380](https://orcid.org/0000-0001-9051-9380)

[†]Senior authors

Damian J Mole[†]

Centre for Inflammation Research, Institute for Regeneration and Repair, Edinburgh Bioquarter, The University of Edinburgh, Edinburgh, UK, Clinical Surgery, The University of Edinburgh, Edinburgh, UK
ORCID iD: [0000-0001-6884-7302](https://orcid.org/0000-0001-6884-7302)

For correspondence: damian.mole@ed.ac.uk

[†]Senior authors

Editors

Reviewing Editor

Omowumi Kayode

Mountain Top University, Makogi Oba, Nigeria

Senior Editor

Benoît Kornmann

University of Oxford, Oxford, United Kingdom

Reviewer #1 (Public review):

Summary:

This study serves as a proof of concept for KMO inhibition as a new non-hormonal treatment for endometriosis. The authors investigated KMO expression in human endometrial and endometriosis lesion tissues, confirmed that KNS898 effectively inhibits KMO and alleviates manifestations of endometriosis in mice - reduced endometriosis lesions and improved hyperalgesia and cage behaviour.

Strengths:

- (1) Inhibition of KMO may present as a promising first-in-class non-hormonal therapeutic agent for patients suffering from endometriosis and the side-effects of hormonal treatments.
- (2) The expression of KMO in endometrial tissues was demonstrated in both human (multiple patients per AFS stage of disease) and mice tissues.
- (3) Measurement of multiple substrates/analytes of the KMO regulatory pathway was performed and demonstrated strong correlation to each other in response to KMO inhibition.
- (4) The aims of study (as proof-of-concept) were achieved in the study and the results support their conclusions.

Weaknesses:

If any dysregulation in the KMO/tryptophan metabolic activity, expression and/or pathway in endometriosis can be shown, this will strengthen the rationale for the use of KMO inhibitor in the disease.

<https://doi.org/10.7554/eLife.99226.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors aim to address the clinical challenge of treating endometriosis, a debilitating condition with limited and often ineffective treatment options. They propose that inhibiting KMO could be a novel non-hormonal therapeutic approach. Their study focuses on:

- Obtaining proof-of-concept for KMO inhibition as a novel therapy for endometriosis.
- Characterising KMO expression in human and mouse endometriosis tissues.
- Demonstrating the efficacy of KMO inhibition in improving histological and symptomatic features of endometriosis.

Strengths:

- **Novelty and Relevance:** The study addresses a significant clinical need for better endometriosis treatments and explores a novel therapeutic target.

Weaknesses:

- **Limited Mechanistic Insight:** The study lacks a comprehensive investigation of the mechanistic pathways through which KNS898 affects endometriosis. The dysregulation of KMO activity and the kynurenine pathway in endometriosis remains poorly characterized, both in the human condition and the experimental model. While the authors present preliminary evidence that kynurenine metabolites (KYN, 3HK, and KYNA) are not dysregulated in the experimental model of endometriosis, they show that KMO inhibition modulates these metabolite levels and leads to some improvement in disease features. However, these findings do not significantly close the existing knowledge gap or provide a strong rationale for targeting KMO as a therapeutic approach for endometriosis. Further mechanistic insights are necessary to justify the potential of KMO inhibition in this context.

Achievement of Aims:

- The authors demonstrated that KMO is expressed in endometriosis lesions and that KNS898 can induce KMO inhibition, leading to biochemical changes and improvements in few endometriosis features in a mouse model. Therefore, the authors addressed the proposed specific aims. However, fail to provide a clear rationale for proposing KMO inhibition as a novel therapy for endometriosis.

Support of Conclusions:

- The conclusions are somewhat overextended given the limitations in mechanistic insights to explain how KMO inhibition result in improvment of histological and symptomatic features of experimental endometriosis. The study provides promising initial evidence but requires further exploration to firmly establish the efficacy of KNS898 for endometriosis treatment.

Impact on the Field:

- The study introduces a novel therapeutic target to be explored for endometriosis, potentially leading to non-hormonal treatment options.

Utility of Methods and Data:

- The methods used provide a foundation for further research, although they require refinement. The data, while promising, need more rigorous investigation and deeper mechanistic exploration to be fully convincing and useful to the community.

<https://doi.org/10.7554/eLife.99226.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This study explores the therapeutic potential of KMO inhibition in endometriosis, a condition with limited treatment options.

Strengths:

KNS898 is a novel specific KMO inhibitor and is orally bioavailable, providing a convenient and non-hormonal treatment option for endometriosis. The promising efficacy of KNS898 was demonstrated in a relevant preclinical mouse model of endometriosis with pathological and behavioural assessments performed.

Weaknesses:

(1) The expression of KMO in human normal endometrium and endometrial lesions was not quantified. Western blot or quantification of IHC images will provide valuable insight.

Given the differential expression of KMO in luminal epithelial cells lining the endometrial glands compared to the other parts of the endometrium, a general endometrial Western Blot prep is not going to be additionally helpful or accurate in addressing this question, without e.g. laser capture microdissection or single cell quantitative proteomics. Furthermore, KMO is a flavin-dependent monooxygenase and the activity, especially generating the oxidative stressor product 3-hydroxykynurenine is far more dependent on kynurenine substrate

availability than it is on actual enzyme abundance - although it is important to show (as we have done), that KMO is present in the human endometrial glands and in human distended endometrial gland-like structures (DEGLS).

If KMO is not overexpressed in diseased tissues i.e. it may have homeostatic roles, and inhibition of KMO may have consequences on general human health and wellbeing.

KMO certainly does have important homeostatic roles, for example as key step in the repletion of NAD⁺ through *de novo* synthesis. Although with good nutrition and sufficient NAD⁺ precursors in the diet e.g. niacin, that specific role may be partially redundant. KMO knockout mice exhibit normal fertility and fecundity and do not show a survival deficit compared to littermate wildtype controls (e.g. Mole et al *Nature Medicine* 2016). To further develop KNS898 towards clinical use, preclinical GLP safety and toxicology studies and human Phase 1 clinical trials will of course need to be completed, but that is standard for the development of any new drug

In addition, KMO expression in control mice was not shown or quantified.

Control mice that were not inoculated intraperitoneally with endometrial fragments did not develop DEGLS and therefore there is nothing to show or quantify.

Images of KMO expression in endometriosis mice with treatments should be shown in Figure 4.

We have now included a representative KMO immunohistochemistry image from each endometriosis group and included all KMO immunohistochemistry images in Supplementary Information.

The images showing quantification analysis (Figure 4A-F) can be moved to supplementary material.

This recommendation contradicts the emphasis placed by the same reviewer earlier regarding quantification, so we have elected to keep it where it is.

(2) Figure 1 only showed representative images from a few patients. A description of whether KMO expression varies between patients and whether it correlates with AFS stages/disease severity will be helpful. Images from additional patients can be provided in supplementary material.

We have added extra information to the Figure legend to clarify the disease stage of the superficial peritoneal lesions which were illustrated (Stage I/II) and to link them to the information in supplementary Table S1. In total we examined 11 peritoneal lesions and 5 ovarian lesions (stage III/IV) – in every sample examined immunopositive staining was most intense in epithelial cells lining gland-like structures. Sections illustrated were chosen to illustrate this key finding.

(3) For Home Cage Analysis, different measurements were performed as stated in methods including total moving distance, total moving time, moving speed, isolation/separation distance, isolated time, peripheral time, peripheral distance, in centre zones time, in centre zones distance, climbing time, and body temperature. However, only the finding for peripheral distance was reported in the manuscript.

This was indeed a large amount of output, which we rationalised for the benefit of a concise paper. The paper now includes a description of which parameters showed a difference with

drug treatment.

(4) The rationale for choosing the different dose levels of KNS898 - 0.01-25mg/kg was not provided. What is the IC50 of a drug?

KNS898 dosing has been extensively characterised by us in multiple species, and the pIC50 has already been published (e.g. Hayes et al Cell Reports 2023 and elsewhere). We now include the pIC50 in the present manuscript to save the reader from having to search through another reference.

(5) Statistical significance:

(a) Were stats performed for Fig 3B-E?

Now included, thank you.

(b) Line 141 - 'P = 0.004 for DEGLS per group'

However, statistics were not shown in the figure.

Thanks, now displayed on figure.

(c) Line 166 - 'the mechanical allodynia threshold in the hind paw was statistically significantly lower compared to baseline for the group'

However, statistics were not shown in the figure.

(d) Line 170 - 'Two-way ANOVA, Group effect P = 0.003, time effect P < 0.0001' The stats need to be annotated appropriately in Figure 5A as two separate symbols.

Arguably the far more important comparison in this figure is whether there is any effect of treatment, and to mark multiple statistical comparisons on the figure would make it difficult to understand. Instead, the figure legend and results text have been clarified on this point.

(e) Figure 5B - multiple comparisons of two-way ANOVA are needed. G4 does not look different to G3 at D42.

Multiple comparison testing (Dunnett's T3) was done and the results have been clarified in the text and figure legends.

*(f) Line 565 - 'non-significant improvement in KNS898 treated groups'. However, ** was annotated in Figure 5A.*

Thank you. This is an error that has been checked and corrected.

(6) Discussion is very light. No reference to previous publications was made in the discussion. Discussion on potential mechanistic pathways of KYR/KMO in the pathogenesis of endometriosis will be helpful, as the expression and function of KMO and/or other metabolites in endometrial-related conditions.

The discussion is deliberately concise and focussed. The paper has 21 references to previous publications. A speculative discussion is generally not favoured by us.

The findings in this study generally support the conclusion although some key data which strengthen the conclusion eg quantification of KMO in normal and diseased tissue is

lacking.

We differ from the reviewer here and do not think that those data would materially affect the likelihood of KMO inhibition being efficacious in human endometriosis in Phase 2/3 clinical trials.

Before KMO inhibitors can be used for endometriosis, the function of KMO in the context of endometriosis should be explored eg KMO knockout mice should be studied.

We take the view that before KMO inhibitors can be used for endometriosis in patients there are multiple other regulatory and clinical development steps that are required that would be a priority. While using a KMO knockout mouse might be an interesting scientific experiment, it would not impact on the critical path in a material way.

Reviewer #2 (Public Review):

Summary:

The authors aim to address the clinical challenge of treating endometriosis, a debilitating condition with limited and often ineffective treatment options. They propose that inhibiting KMO could be a novel non-hormonal therapeutic approach. Their study focuses on:

- Characterising KMO expression in human and mouse endometriosis tissues.*
- Investigating the effects of KMO inhibitor KNS898 on inflammation, lesion volume, and pain in a mouse model of endometriosis.*
- Demonstrating the efficacy of KMO blockade in improving histological and symptomatic features of endometriosis.*

Strengths:

- Novelty and Relevance: The study addresses a significant clinical need for better endometriosis treatments and explores a novel therapeutic target.*
- Comprehensive Approach: The authors use both human biobanked tissues and a mouse model to study KMO expression and the effects of its inhibition.*
- Clear Biochemical Outcomes: The administration of KNS898 reliably induced KMO blockade, leading to measurable biochemical changes (increased kynurenine, increased kynurenic acid, reduced 3-hydroxykynurenine).*

Weaknesses:

- Limited Mechanistic Insight: The study does not thoroughly investigate the mechanistic pathways through which KNS898 affects endometriosis. Specifically, the local vs. systemic effects of KMO inhibition are not well differentiated.*

While we agree that this is not a comprehensive mechanistic analysis, given that the ultimate therapy would be almost certainly a once daily oral dosing i.e. systemic administration, we do not consider differentiating local vs systemic effects of KMO inhibition to be critical to therapeutic development in this scenario.

- Statistical Analysis Issues: The choice of statistical tests (e.g., two-way ANOVA instead of repeated measures ANOVA for behavioral data) may not be the most appropriate, potentially impacting the validity of the results.*

The selection of two-way ANOVA (time and group) is sufficient and correct for this experimental analysis and its use does not invalidate the results. We agree that repeated measures ANOVA could be a valid alternative.

- *Quantification and Comparisons: There is insufficient quantitative comparison of KMO expression levels between normal endometrium and endometriosis lesions,*

Please see response above to quantification question raised by Reviewer 1.

- and the systemic effects of KNS898 are not fully explored or quantified in various tissues.*

Please see earlier responses. KNS898 has been thoroughly explored in multiple tissues, species and experimental models, but those data do not need rehearsed here.

- *Potential Side Effects: The systemic accumulation of kynurenine pathway metabolites raises concerns about potential side effects, which are not addressed in the study.*

As discussed above (response to Reviewer 1), KMO knockout mice exhibit normal fertility and fecundity and do not show a survival deficit compared to littermate wildtype controls (e.g. Mole et al *Nature Medicine* 2016). To further develop KNS898 towards clinical use, preclinical GLP safety and toxicology studies and human Phase 1 clinical trials will naturally need to be completed, but this is standard for the development of any new drug.

Achievement of Aims:

- *The authors successfully demonstrated that KMO is expressed in endometriosis lesions and that KNS898 can induce KMO blockade, leading to biochemical changes and improvements in endometriosis symptoms in a mouse model.*

Support of Conclusions:

- *While the data supports the potential of KMO inhibition as a therapeutic strategy, the conclusions are somewhat overextended given the limitations in mechanistic insights and statistical analysis. The study provides promising initial evidence but requires further exploration to firmly establish the efficacy and safety of KNS898 for endometriosis treatment.*

We do not agree that the conclusions are overextended based on the data presented, as expanded in the reply to the eLife editorial assessment at the beginning of this response. It is clear that additional preclinical, regulatory and clinical development work, and human clinical trials will be required to firmly establish the efficacy and safety of KN898 for endometriosis treatment.

Impact on the Field:

- *The study introduces a novel therapeutic target for endometriosis, potentially leading to non-hormonal treatment options. If validated, KMO inhibition could significantly impact the management of endometriosis.*

Utility of Methods and Data:

- *The methods used provide a foundation for further research, although they require refinement. The data, while promising, need more rigorous statistical analysis and deeper mechanistic exploration to be fully convincing and useful to the community.*

We believe that the data are a) convincing, and b) useful to the community. To be advanced effectively towards patients, KNS898 needs to follow the critical development path outlined above.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Change 'hyperalgia' to hyperalgesia throughout the manuscript including the title.

Done

(2) Line 69 - write '3-HK' in full.

Done

(3) Line 85 - the findings of the study include 'define the preclinical efficacy of KNS898 in reducing inflammation'. The inflammatory profile was not studied.

Changed to "disease"

(4) Line 259 - write 'EPHect' in full.

Done

(5) Line 260 - write 'AFS' in full. Also, abbreviate 'AFS' in the caption of Table S1.

Done

(6) 20 patients were listed in Table S1 but only 19 were accounted for in the methods section.

Apologies there was an error and has now been corrected in the methods section as one of the endometrial samples had not been included. Table S1 has also been changed to make it clear which samples were eutopic endometrium to differentiate them from the lesions.

(7) The location from which the endometrial lesion tissues were obtained should be provided in Table S1.

Table S1 has been changed to make it clear that the subtypes of lesions examined were classified as Stage I/II – superficial peritoneal subtype and Stage III/IV – endometrioma. The methods section has also been updated to reflect these subtypes (lines 272-277).

(8) Table S2 - G5 should be given compound 'A' not 'B'.

Thank you. Corrected.

(9) Figure 2E was not referenced in the text and no figure legend was provided.

Now referenced and the figure legend updated.

(10) Figure 3A - font needs to be enlarged. HCA baseline recording was annotated as performed twice in the protocol. When is the baseline taken and on what day was the Week 12 measurement taken (refer to Figures 5C and D)?

Font has been enlarged as requested. The second HCA baseline annotation in Fig 3A is a cut-and-paste error, now rectified and the time of second measurement annotated.

(11) Line 133 - 'In KNS898-treated group G4 (endometriosis + treatment from Day 19), DEGLS formed in 4 of 15 mice (26.7%) and in G5 (Endo + treatment start on Day 26) in 6 of 15 mice (40%) (Fig. 3f)'. The aforementioned data is not reflected in Figure 3F.

Thank you. This has been rectified.

(12) Line 137 - 'Mice with endometriosis receiving KNS898 from the time of inoculation (G4) had an average of 2.0 DEGLS per animal with DEGLS (total = 8 DEGLS in 4 mice in G4) and those receiving KNS898 1 week after inoculation (G5) had an average of 1.8 DEGLS per animal (total = 11 DEGLS in 6 mice in G5) (Figs. 3g and 3h).'

The aforementioned data is not reflected in Figure 3G. There is no Figure 3H shown.

Rectified as above.

(13) Provide a discussion of why KA levels were significantly lower in Figure 3E compared to Figure 2C.

(14) Figure legend for Figure 3 - G1 and G2 were noted as n=8. However, Figure S1 and Table S2 noted both groups as n=10.

Thank you. This is a typographical error. The legend for Fig 3 should indeed read n=10 for G1 and G2 and has been corrected.

(15) Line 181 - 'compared to non-operated and sham-operated control groups'. Only the sham group was shown in Figures 5C and D.

This text has been clarified to refer only to the data shown.

(16) Figure 1 images need scalebars. Same for Figure 4.

Now added

(17) Figure 3B - y-axis is fold change?

Relative concentration. Legend has been clarified.

(18) Figures 5A and B - are the last Von Frey measurements taken on Day 40 (as per Figure 3A) or 42?

Taken on Day 42. Fig 3A (the prospective protocol figure) has been clarified to reflect what actually happened (D42) as opposed to what was planned (D40) to pre-empt any further confusion.

(19) Symbols in Figure S1 need to be explained in the Figure legend.

Done

(20) Figures 2A and 2D should not be plotted in log scale to match the description of results in Line 106 and Line 118.

These particular results are plotted on a log scale to allow the reader to visualise that detectable levels of drug are measurable at very low doses and that there is no significant pharmacodynamic effect at that low dose. We choose to retain the present format.

Reviewer #2 (Recommendations For The Authors):

Comments and queries

Introduction/aims section:

Line 82 - 87: Clarify in the proposal aims what is being accessed and analysed in humans and/or in animal models (mice). Specifically state clearly the correlations with KMO expression. Were the correlations between KMO expression with features of inflammation performed only in mice or also in humans?

Thank you for this comment. The aims have been clarified in the Introduction.

Section - KMO is expressed in human eutopic endometrium and human endometriosis tissue lesions:

Was any quantitative or semi-quantitative method used to quantify the KMO expression in human tissues? Although the authors claimed that "KMO was strongly immunopositive in human peritoneal endometriosis lesions" by the representative figures it is not clear if KMO expression is similar, higher or lower between normal endometrium and peritoneal endometriosis lesions.

We have added extra information to the legend of Figure 1 to identify the PIN number of the superficial lesions illustrated. The key finding from the immunostaining with the antibody which had been previously validated as specific for KMO was that the most intense immunopositive response was in glandular epithelial cells and the samples illustrate this result.

Section - Oral KNS898 inhibits KMO in mice:

The authors clearly confirmed the target engagement of KNS898 in inhibiting KMO activity and, therefore, affecting upstream and downstream metabolites systemically in (peripheral fluid/ plasma) mice. Whether KNS898 effect is broad and targets systemic immune cells and whole body cells and tissue was not explored. It was also not explored if KNS898 is able to specifically inhibit KMO locally at the endometrium tissue by targeting epithelial and/or infiltrated immune cells, for example.

That is correct.

It would be interesting to measure (or if it was measured to report in this section and also in Figure 2) the levels of KYN, KA and 3HK in naïve animals that did not receive KNS898. It would help to understand the net effect of KNS898 on the levels of kynurenine pathway metabolites and, therefore, justify the dose chosen.

These data are already presented in Fig 3B-E, control group.

Perhaps then the chosen dose could be lower considering the possible substantial changes in kynurenine pathway metabolites levels, which are reported to exert an effect in many cells, tissues and systems and could, therefore, precipitate side effects. Even more considering that the values for these metabolites are expressed as ng/ml, which

hinders the comparison of the metabolite levels with the one reported for naïve animals in the literature. I would also suggest expressing the metabolite levels as nM/L.

This is not a relevant method of determining dose-limiting toxicity or safety pharmacology/toxicology, either non-GLP or GLP. There are international guidelines on the proper conduct of those studies. This is also why it is important not to make claims about the safety or otherwise of an experimental compound in an in vivo setting that has not explicitly complied with those regulatory standards. With regard to the units recommendation, accepted units are ng/mL or nM, not usually nM/L.

Section - KMO blockade reduces endometrial gland-like lesion burden in experimental endometriosis in mice:

Line 130: It would be better to replace "blockade of 3HK production" with "reduction of 3HK production" to better reflect the results.

Changed to "inhibition of 3HK production".

Line 140: In G5 (treatment starting at Day 26/ 1 week after inoculation), is the experimental model of endometriosis already established with all pathological and phenotypic features?

This was not specifically tested in this experiment.

Lines 146 - 148: It would be better to specify that "Overall, there was no significant difference IN BODY WEIGHT between G3 and the KNS898 treatment groups G4 and G5 (endometriosis + treatment from Day 26)". Otherwise, this last sentence might be interpreted as the overall conclusion of this result sub-section.

Thank you, a good point and has been corrected.

The authors demonstrated with an experimental approach that KMO blockade reduces a pathological measure of endometriosis i.e., endometrial gland-like lesion burden, in experimental endometriosis in mice when both administrated concomitant but also after the disease development. Although mechanistic insights about how reduced KMO activity can reduce the developed distended endometrial gland-like structures were not explored. Therefore, it remains to be investigated which (and how) kynurenine pathway metabolites are directly linked to the beneficial effects of KMO blockade in the experimental model of endometriosis.

We agree.

Although the beneficial effects on the pathological measures are evident, Figure 3 shows an exorbitant accumulation of KYN and KA and also a substantial reduction in 3HK after the treatment with KNS898, which then raises concerns about tolerability and side effects. Would this effective KNS898 dose be viable and translational as a therapeutic approach?

Please refer to comments above at multiple junctures about safety pharmacology and the clinical development critical path.

Section - KMO is expressed in experimental endometriosis in mice:

By histological examination, the authors confirm that the treatment with KNS898 specifically reduced the KMO expression intensity in the DEGLS from mice. Therefore, the

effect exerted by KNS898 locally on the KMO expression at the DEGLS could be, at least, partially responsible for the beneficial effects observed in Figure 3 i.e., the reduction of pathological measures. Although remains to be explored whether the effect of KNS898 in other cells or tissues could also be accountable for the beneficial effects exerted by KNS898 on the animal model of endometriosis.

This is correct.

From a logical experimental point of view, I would suggest switching the order of the result subsection "KMO blockade reduces endometrial gland-like lesion burden in experimental endometriosis in mice" and "KMO is expressed in experimental endometriosis in mice" as well as the respective Figures 3 and 4.

We do not agree. Fig 3 (and section) is the macroscopic enumeration of DEGLS, Fig 4 (and section) is the microscopic and immunohistochemical evaluation of the lesions introduced in Fig 3. The sequence as originally presented is the more logical.

Sections - KMO inhibition reduces mechanical allodynia in experimental endometriosis - and - KMO inhibition reduces mechanical allodynia in experimental endometriosis:

The authors suggested that the KMO inhibition with KNS898 exerts beneficial effects on behavioural paradigms related to the experimental model of endometriosis. Based on the statistical analysis performed for the author, KMO inhibition with KNS898 reduces mechanical allodynia, as well as rescues, impaired cage exploration behaviour and mobility in mice with endometriosis. However, I believe that the most indicated statistical tests for Von Frey (allodynia behaviour) and Home cage (illness behaviour) analyses over time would be repeated measures ANOVA and paired t-test, respectively (and not two-way ANOVA as performed). Therefore for a more trustful analysis and interpretation of this data set, I would suggest the authors modify the statistical analysis and report the corresponding interpretation of these tests.

The selection of two-way ANOVA (time and group) is suitable for this experimental analysis and its use does not invalidate the results. We agree that repeated measures ANOVA could be a valid alternative.

Overall, the authors present a solid and useful case for KMO inhibition as a potential therapeutic strategy for endometriosis. However, the study would benefit from more detailed mechanistic insights, appropriate statistical analyses, and an evaluation of potential side effects. With these improvements, the research could have a significant impact on the field and pave the way for new treatment modalities for endometriosis.

We thank the reviewer for the positive comments and we have responded to the criticisms above.

Specific recommendations for improvement:

- *Mechanistic Studies: Conduct detailed studies to understand the local vs. systemic effects of KMO inhibition and its specific impacts on different cell types and tissues. If not feasible here, the authors could include in the discussion section a detailed overview of the possible mechanisms implicated.*

While we agree that this is not a comprehensive mechanistic analysis, given that the ultimate therapy would be almost certainly a once daily oral dosing i.e. systemic administration, we do not consider differentiating local vs systemic effects of KMO inhibition to be critical to therapeutic development in this scenario. We do not think speculation about possible

mechanisms that is not supported by experimental data should be included. Furthermore, that notion (of statements not supported by data) has been given as a criticism by the reviewers, and therefore consistency on this point must be preferable.

- *Quantitative Analysis: Include more robust quantitative methods to compare KMO expression levels in different tissues and assess the correlation between KMO expression and pathological and behavioural changes.*

As discussed above, the pathophysiological importance of KMO is in its enzymatic activity, not in its abundance as a protein, and 3HK production is far more dependent on kynurenine substrate availability rather than KMO protein abundance.

- *Appropriate Statistics: Use the most suitable statistical tests for behavioural and other repeated measures data to ensure accurate interpretation.*

As discussed above

- *Side Effect Evaluation: Investigate potential side effects of systemic KMO inhibition, particularly focusing on the long-term implications of altered kynurenine pathway metabolites. If not feasible here, the authors could include in the discussion section a detailed overview of the possible side effects associated as well as inform if KNS898 can cross the BBB and its implications.*

For a novel small molecule therapeutic compound in preclinical/clinical development, there are strictly regulated preclinical and clinical development standards that need to be met. It would not be responsible to publish or make claims about safety and potential adverse effect profiles without conducting the proper panel of tests within a suitable regulatory framework.

<https://doi.org/10.7554/eLife.99226.2.sa0>